Document 5 — PPT Creation Guide

SIH26166 · Slide-by-slide structure, content, and design notes

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Slide-by-slide structure (10 slides, ~5 minutes)

Slide 1 — Title

- PS number (SIH26166), one-line problem restatement in your own words, team name, team members.
- Design: your terminator-disc motif from the workplan (a half-lit lunar disc) works well as a recurring visual anchor across every slide's corner — keep it small and consistent, not decorative-heavy.

Slide 2 — The stakes (~30s)

- One sentence: multi-instrument correspondence is the bottleneck under mosaicking, DEM generation, and landing-site selection.
- One supporting fact: ISRO's own OHRC-based landing-site pipeline (OPTIMUS) isn't publicly documented — this is a real, named gap in public tooling, not an invented one.
- Visual: a simple 3-instrument comparison graphic (OHRC/TMC-2/IIRS footprint sizes to scale) — this alone communicates "these are wildly different sensors" faster than a sentence.

Slide 3 — The named gap (~30s)

- State the ISRO SAC (Makharia et al., 2025) benchmark finding directly, in your own words: classical feature matching (SIFT/ASIFT) achieves sub-pixel accuracy in good conditions but breaks down under shadowing and subtle intensity shifts.
- Put the citation on the slide, small, bottom corner — verified DOI/title, exactly as it exists in the source. This single line is your strongest credibility signal in the whole deck.

Slide 4 — Your one improvement (~60s)

- Name the lane you took plainly: e.g. "a shadow-invariant learned descriptor, trained on DEM-derived synthetic sun-angle pairs, used alongside classical SIFT."
- Do **not** claim to have solved multi-modal registration generally — state the one failure mode you targeted.
- One diagram: source patch → descriptor network → embedding distance → candidate match, feeding the same RANSAC verification as the classical branch.

Slide 5 — The math, one slide (~20s)

- Pick exactly one equation you can defend under questioning — the contrastive/triplet loss, or your Fourier-surface truncation, or the RANSAC iteration-count formula. Show it, and one sentence on what it buys you.
- Don't try to fit five equations on one slide — one, explained well, beats five, explained badly.

Slide 6 — The demo (~60s)

- Two real scenes, visibly different shadow patterns, side by side.
- Left: classical baseline's matches visibly scattering/failing under shadow.
- Right: your method's matches holding.
- Beneath both: the R3F terrain view with contour lines, so judges see a working system, not a scatter plot.

- If live-demoing, have a recorded backup video ready — Wi-Fi at a venue is not something to bet a live pipeline call on.

Slide 7 — Measured results (part of the demo, or its own slide if numbers are rich enough)

- A small table: RMSE, inlier count, inlier ratio, for baseline vs. your method, on your real scene pair.
- If you also ran the synthetic DEM-pair validation, show that number alongside, labeled clearly as synthetic (never blur the line between measured-on-real-data and measured-on-synthetic-data).

Slide 8 — Honest scope (~30s)

- Name the one failure mode you focused on, out loud, and name 2–3 things you deliberately left out and why (e.g. full Hapke inversion, real IIRS cross-modal matching as core rather than stretch).
- This slide is a credibility asset, not a weakness — frame it that way in your delivery, not apologetically.

Slide 9 — Architecture

- The system diagram (frontend / Supabase / FastAPI+ONNX / Upstash / Sentry / Gemini) from your build plan. Judges scoring "is this a working system" want to see this, not just the pipeline math.

Slide 10 — Close

- One line restating the actual claim: a working, generic, evaluable registration pipeline across OHRC/TMC-2/IIRS against external reference imagery, with one measured, named improvement.
- Thank-you + contact/repo link.

Design notes

- **Consistency over decoration.** Reuse one accent color scheme and one typeface pairing across all 10 slides (your workplan HTML already has a coherent palette — amber/cyan/violet on a warm neutral background — carry that into the deck rather than inventing a new one).
- **One idea per slide.** If a slide needs two sentences to explain what it's showing, it's two slides.
- **Never put a number on a slide you can't defend.** Every RMSE, inlier count, or percentage on the deck should trace back to an actual run you can point to in your repo/logs.
- **Rehearse the timing**, not just the content — 5 minutes disappears fast once Q&A framing eats into it; know which slides you'd cut first if running long (Slide 7 is the safest one to compress into Slide 6 if needed).

Building it

Fastest path for a hackathon timeline: build this as a Google Slides or PowerPoint deck directly from this outline, or hand this document's slide breakdown to an AI drafting tool (Gemini, or Claude for PowerPoint) as structured input — feed it your actual measured numbers once you have them, not placeholder figures, and treat the AI's phrasing as a first draft to tighten, not a finished pitch.